

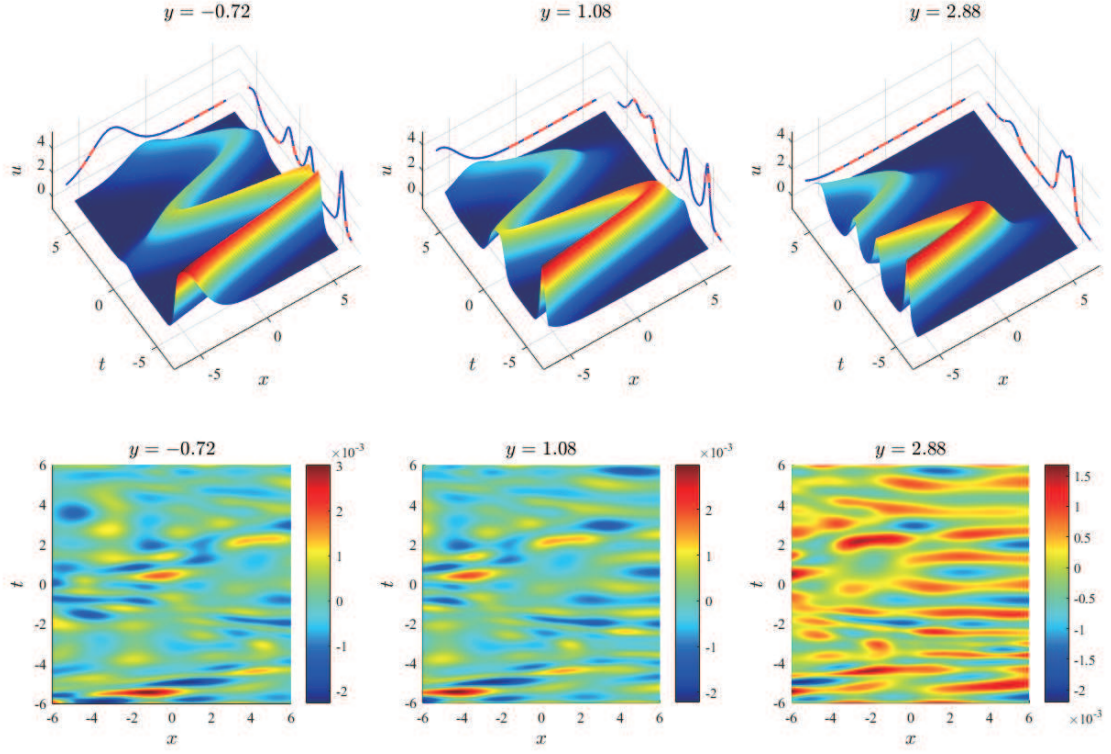


Figure 17: (Color online) Data-driven discovery of the nonlinear coefficient $v(t) = \cos(t)$ in the variable coefficient KP equation: the upper part of the figure displays the 3D predicted solution at these three cross-sections. The red dashed lines represent the predicted solution, while the blue lines represent the true solution; the lower part displays the errors between the predicted and true solutions at the same cross-sections.

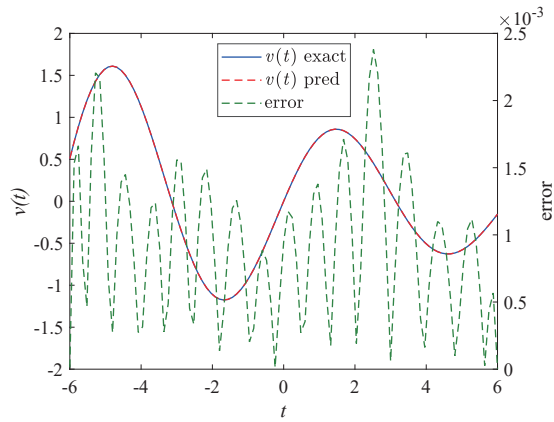


Figure 18: (Color online) The data-driven variable coefficient $V(t)$ discovery of the variable coefficient KP equation by R-gPINN related to solution (21): the comparison of the error between the learning and exact variable coefficient $V(t)$ over the interval $[-6, 6]$.